Food and body-image themes in annotated social media posts

INFO 2950 · Phase II · Parts 1 and 2
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Github repo: AnyaChoudhary/info2950-fall2026-project

1. Research questions

Central question: How are food- and body-related concerns represented, and how do they relate to other content features within two annotated social-media samples?

Our original motivation concerned social media, food attitudes, body image, food guilt, social comparison, recovery, and diet culture. The available public files support a narrower **content-level** investigation:

1. What concerns and content features are recorded within each dataset, and how common are they among the posts eligible for each annotation?

2. Among Twitter posts coded as ED/body-image related, is a shape-concern annotation associated with an eating-concern annotation?

3. Within EDTT, is a restraint annotation associated with audio valence after accounting for post format?

Questions 2 and 3 are reserved for the final report. Phase II uses marginal summaries and different exploratory relationships to understand the data. We do not estimate algorithm effects, exposure, engagement effects, diagnoses, or changes in an individual's attitudes. Recovery and social comparison are not consistently measured in both public files and are not treated as harmonized outcomes.

Why two sources? Twitter supplies a larger coded text/image sample; EDTT supplies a smaller multimodal sample. They offer complementary descriptions. We do not pool rows, compare their percentages as platform effects, or treat them as a continuous time series.

2. Data collection and cleaning

2.1 Collection and provenance

The source files were downloaded on October 15, 2026. Downloads occurred outside the notebook, as the rubric requests collection descriptions but no collection code. Original bytes are retained under `data/raw/`; the manifest records URLs and SHA-256 checksums. The same unchanged files are hosted in the project GitHub raw-data folder, pinned to the verified upload commit so reviewers can retrieve the exact inputs.

• **Twitter:** Cavazos-Rehg et al. (2019), PLOS ONE article, S1 Dataset, and S1 Codebook. The study sampled 3,000 English-language posts from a keyword-selected January 2015 corpus. The release supplies de-identified codes. The original post text is not included. The published methods describe trained coders and adjudication.

• **TikTok:** Shaveet, Viranda, and Choudhury (2026), EDTT v2, public CSV, and data dictionary. This is a curated, expert-annotated pro-ED content sample assembled through text retrieval and recommendation surfacing. Private post identifiers and transcripts are withheld. The record was published in April 2026; the proposed 2024–2025 collection window was not verified in the public documentation inspected and is not assumed here.

• **Existing project:** GitHub repository, inspected at commit `1c88711a7ee4e535677e01e2d5a6ac09674efa0c`. At that initial inspection its root contained only `.gitignore`; no prior notebook or Phase I submission was available to reuse. The source files and provenance manifest have since been added at commit `905e574351e821300dc5b199747216a3bd630e1b`; each uploaded file was read back and

matched to the local file.

No scraping, individual contact, account identification, or new annotation was performed. No API credentials were required.

2.2 Environment and integrity checks

Run this notebook with the included data/ directory. It locates the project root automatically when opened from the repository's notebooks/ folder or the extracted bundle. The next cell loads libraries, checks file integrity, and configures consistent figures. A changed raw file fails explicitly rather than silently changing the results.

Code cell [1]

```
# Initialize the analysis and verify that the supplied source files are unchanged.
from pathlib import Path
import hashlib, json, re, sys, platform
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

# Find data from either the bundle root or the repository's notebooks folder.
ROOT = next((p for p in [Path.cwd(), *Path.cwd().parents]
             if (p / 'data/raw/manifest.json').exists()), None)
assert ROOT is not None, 'Keep this notebook with the supplied data directory.'
RAW = ROOT / 'data/raw'
INTER = ROOT / 'data/intermediate'
CLEAN = ROOT / 'data/processed'
FIG = ROOT / 'figures'
for folder in [INTER, CLEAN, FIG]:
    folder.mkdir(parents=True, exist_ok=True)
manifest = json.loads((RAW / 'manifest.json').read_text())
for item in manifest:
    actual = hashlib.sha256((RAW / item['file']).read_bytes()).hexdigest()
    assert actual == item['sha256'], item['file']
assert hashlib.md5((RAW / 'EDTT_public.csv').read_bytes()).hexdigest() ==
'8bf6f87fd0ff4147790209f66ee364ea'
# Prefer Times New Roman; keep a portable serif fallback for other computers.
from matplotlib import font_manager
font_folder = Path('/System/Library/Fonts/Supplemental')
for font_file in font_folder.glob('Times New Roman*.ttf'):
    font_manager.fontManager.addfont(str(font_file))
plt.rcParams.update({'font.family': 'serif',
                     'font.serif': ['Times New Roman', 'Times', 'DejaVu Serif'],
                     'text.color': 'black', 'axes.labelcolor': 'black',
                     'axes.titlecolor': 'black', 'xtick.color': 'black',
                     'ytick.color': 'black', 'figure.dpi': 130, 'savefig.dpi': 180,
                     'font.size': 10, 'axes.spines.top': False,
                     'axes.spines.right': False, 'axes.titleweight': 'bold'})
BLUE, TEAL, ORANGE = '#315B86', '#16847D', '#C87A36'
def finish(name):
    plt.savefig(FIG / (name + '.png'), bbox_inches='tight')
    plt.show()
print('All four source checksums verified; EDTT also matches Zenodo MD5.')
print(f'Python {platform.python_version()} | pandas {pd.__version__} | '
      f'numpy {np.__version__} | matplotlib {matplotlib.__version__}')
```

```
All four source checksums verified; EDTT also matches Zenodo MD5.
Python 3.12.0 | pandas 3.0.6 | numpy 2.5.4 | matplotlib 3.11.2
```

2.3 Twitter cleaning: respect the codebook's eligible populations

Each row has a synthetic id, not a tweet ID. We keep all 3,000 rows in the validated intermediate file, use nullable integer codes, and retain blank conditional fields. ed_about=9 means unknown, not yes. Only ed_about=1 enters the main analysis file.

The codebook defines three kinds of subpopulation: shape-concern posts for image availability, shape-concern posts with an image for visual attributes, and eating-concern posts for restraint/purge. Aversion is assessed only when none of the three concerns is present. The next cell checks every field against these eligibility rules. It stops on an unexpected code, duplicate ID, or missing value inside an eligible population. Repeated annotation patterns are retained because distinct posts can share codes.

Code cell [2]

```
# Validate every Twitter code and its conditional eligibility before filtering.
tw_raw = pd.read_csv(RAW / 'twitter_s1.csv')
tw = tw_raw.copy()
tw.columns = tw.columns.str.strip().str.lower()
assert tw['id'].notna().all() and tw['id'].is_unique
for col in tw:
    tw[col] = pd.to_numeric(tw[col], errors='raise').astype('Int64')
    if col != 'id':
        allowed = {0, 1, 9} if col == 'ed_about' else {0, 1}
        assert set(tw[col].dropna()).issubset(allowed), col
eligible = {'ed_about': pd.Series(True, index=tw.index)}
for col in ['shape', 'eating', 'weight', 'noconcern']:
    eligible[col] = tw['ed_about'].eq(1)
eligible['noimage'] = tw['shape'].eq(1).fillna(False)
for col in ['body', 'thighs', 'stomach', 'hipbones', 'ribs', 'collarbones', 'extreme']:
    eligible[col] = (tw['shape'].eq(1) & tw['noimage'].eq(0)).fillna(False)
for col in ['restraint', 'purge']:
    eligible[col] = tw['eating'].eq(1).fillna(False)
eligible['aversion'] = tw['noconcern'].eq(1).fillna(False)
audit = []
for col, mask in eligible.items():
    inside_missing = int(tw.loc[mask, col].isna().sum())
    outside_recorded = int(tw.loc[~mask, col].notna().sum())
    assert inside_missing == 0 and outside_recorded == 0, col
    audit.append([col, int(mask.sum()), int((~mask).sum()), inside_missing])
tw.to_csv(INTER / 'twitter_validated.csv', index=False)
twitter = tw.loc[tw['ed_about'].eq(1)].copy()
# Audit a logical inconsistency without overwriting any original annotation.
expected_none = twitter[['shape', 'eating', 'weight']].eq(0).all(axis=1).astype(int)
logic_mismatch = twitter['noconcern'].ne(expected_none)
twitter['noconcern_inconsistent'] = logic_mismatch.astype('Int64')
twitter.loc[logic_mismatch, ['id', 'noconcern_inconsistent']].to_csv(
    CLEAN / 'twitter_logic_flags.csv', index=False)
print('Rows with inconsistent noconcern code:', int(logic_mismatch.sum()))
twitter.to_csv(CLEAN / 'twitter_related.csv', index=False)
twitter_audit = pd.DataFrame(audit, columns=[
    'field', 'eligible n', 'outside universe n', 'unexpected missing n'])
display(twitter_audit)
display(tw['ed_about'].value_counts().rename_axis('ed_about').to_frame('n'))
print('Unique IDs:', tw.id.nunique(), '| Full-row duplicates:', tw.duplicated().sum())
print('Retained related posts:', len(twitter), '| Excluded unrelated/unknown:', len(tw)-len(twitter))
```

Rows with inconsistent noconcern code: 11

	field	eligible n	outside universe n	unexpected missing n
0	ed_about	3000	0	0
1	shape	2584	416	0
2	eating	2584	416	0
3	weight	2584	416	0
4	noconcern	2584	416	0
5	noimage	1679	1321	0
6	body	861	2139	0

	field	eligible n	outside universe n	unexpected missing n
7	thighs	861	2139	0
8	stomach	861	2139	0
9	hipbones	861	2139	0
10	ribs	861	2139	0
11	collarbones	861	2139	0
12	extreme	861	2139	0
13	restraint	326	2674	0
14	purge	326	2674	0
15	aversion	599	2401	0

	n
ed_about	
1	2584
0	342
9	74

Unique IDs: 3000 | Full-row duplicates: 0
Retained related posts: 2584 | Excluded unrelated/unknown: 416

2.4 TikTok cleaning: parse category headers without inventing labels

The public EDTT CSV has no post ID. We assign record_id from the original row order solely to track this file. We check exact duplicates but do not infer duplicate posts from similar summaries or shared audio.

We standardize column names, trim surrounding whitespace, preserve annotation wording, validate numeric ranges and post format, and derive indicators from explicit category headers. Header matching is case-insensitive and allows a header after either a semicolon or a colon; this accommodates an observed separator inconsistency in the visual annotations. We do not silently fix item-level spelling errors or reinterpret free text.

A blank EDE subscale field remains unknown for **all four** derived EDE indicators; it is not recoded to four zeros. A zero on a nonblank field means the category was not recorded there, not that a clinical symptom is absent. Blank diagnostic fields are also preserved; their meaning is not resolved by the inspected dictionary. Numeric audio labels are checked against the dictionary's thresholds, and disagreements are reported rather than overwritten.

Code cell [3]

```
# Normalize the EDTT schema while retaining raw strings and missingness.
tt_raw = pd.read_csv(RAW / 'EDTT_public.csv')
tt = tt_raw.copy()
tt.columns = [re.sub(r'[^a-z0-9]+', '_', c.lower()).strip('_') for c in tt.columns]
for col in tt.select_dtypes(include=['object', 'string']):
    tt[col] = tt[col].astype('string').str.strip().replace('', pd.NA)
tt.insert(0, 'record_id', np.arange(1, len(tt)+1))
assert tt['type'].isin(['video', 'photos']).all()
nums = ['audio_tempo_bpm', 'audio_valence', 'audio_arousal', 'harmonic_ratio', 'mode_major']
for col in nums:
    tt[col] = pd.to_numeric(tt[col], errors='raise')
assert tt[nums].notna().all().all()
```

```
assert tt['audio_tempo_bpm'].gt(0).all()
for col in ['audio_valence', 'audio_arousal', 'harmonic_ratio']:
    assert tt[col].between(0, 1).all(), col
assert tt['mode_major'].isin([0, 1]).all()
tt.to_csv(INTER / 'edtt_normalized.csv', index=False)

def header_flag(series, header):
    pattern = r'(?:^|[;:])\s*' + re.escape(header) + r'\s*:'
    return series.str.contains(pattern, case=False, regex=True).astype('Int64')

ede_headers = {'shape': 'Shape Concern', 'eating': 'Eating Concern',
               'weight': 'Weight Concern', 'restraint': 'Restraint'}
visual_headers = {'body_checking': 'Body Checking & Self Emphasis',
 'on_screen_text': 'On-Screen Text', 'performance': 'Performance & Body Language',
 'weight_imagery': 'Weight-Related Imagery', 'food_imagery': 'Food-Related Imagery',
 'media_memes': 'Media & Memes', 'summaries_metrics': 'Summaries & Metrics'}
for key, header in ede_headers.items():
    tt['ede_' + key] = header_flag(tt['ede_subscale_items'], header)
for key, header in visual_headers.items():
    tt['visual_' + key] = header_flag(tt['visual_elements'], header)
ede_cols = ['ede_' + k for k in ede_headers]
visual_cols = ['visual_' + k for k in visual_headers]
# Ensure all nonblank annotations contain recognized categories.
assert tt.loc[tt.ede_subscale_items.notna(), ede_cols].sum(axis=1).gt(0).all()
assert tt[visual_cols].sum(axis=1).gt(0).all()
assert tt.loc[tt.ede_subscale_items.isna(), ede_cols].isna().all().all()
tt['ede_category_count'] = tt[ede_cols].sum(axis=1, min_count=4).astype('Int64')
label_audit = []
for col in ['audio_valence', 'audio_arousal']:
    expected = pd.Series(np.select([tt[col] < .33, tt[col] <= .66],
                                   ['low', 'medium'], default='high'), index=tt.index)
    label_audit.append([col, int(expected.ne(tt[col + '_label']).sum())])
tiktok = tt.copy()
tiktok.to_csv(CLEAN / 'edtt_features.csv', index=False)
display(pd.DataFrame(label_audit, columns=['audio field', 'label disagreements']))
display(tt_raw.isna().sum().rename_axis('raw field').to_frame('blank n'))
print('EDTT rows retained:', len(tiktok), '| Exact raw duplicates:', tt_raw.duplicated().sum())
print('Known EDE annotation:', tt.ede_subscale_items.notna().sum(),
      '| Blank EDE annotation:', tt.ede_subscale_items.isna().sum())
```

	audio field	label disagreements
0	audio_valence	0
1	audio_arousal	0

raw field	blank n
Type	0
Summary	0
EDE Subscale Items	6
EDE Diagnostic Items	58
Visual Elements	0
Audio Elements	0
Audio Tempo (bpm)	0
Audio Valence	0
Audio Valence Label	0

	blank n
Audio Arousal	0
Audio Arousal Label	0
Harmonic Ratio	0
Mode Major	0

EDTT rows retained: 85 | Exact raw duplicates: 0
Known EDE annotation: 79 | Blank EDE annotation: 6

2.5 Cleaning decisions and saved outputs

The cleaning produces two separate analysis tables. No merge is performed because there is no shared observation key or equivalent sampling design. **Source inconsistency:** 11 Twitter rows have noconcern=0 while all three concern fields are zero. We flag these rows in noconcern_inconsistent, retain the original codes, and save their synthetic IDs separately. The aversion denominator continues to follow the original source eligibility code, not a reconstructed universe. Without the original content we cannot adjudicate which field is wrong. No rows are removed from EDTT; Twitter excludes only the 342 unrelated and 74 unknown-relevance rows from its main analysis. We do not impute values, remove outliers, deduplicate identical annotation patterns, or recode structural blanks to zero.

Intermediate files preserve the complete validated Twitter release and normalized EDTT release. Processed files preserve the variables needed for future analysis. The following audit documents sizes and retention; any future change to exclusions should be recorded as an amendment.

Code cell [4]

```
# Record row counts and retention decisions in a machine-readable audit.
flow = pd.DataFrame([
 ['Twitter', len(tw_raw), len(twitter), len(tw_raw)-len(twitter), 'ed_about must equal 1'],
 ['EDTT', len(tt_raw), len(tiktok), 0, 'retain all public rows']
], columns=['source', 'raw n', 'analysis n', 'excluded n', 'rule'])
flow.to_csv(CLEAN / 'row_flow.csv', index=False)
twitter_audit.to_csv(CLEAN / 'twitter_eligibility_audit.csv', index=False)
display(flow)
print('Analysis table dimensions:', twitter.shape, 'and', tiktok.shape)
```

	source	raw n	analysis n	excluded n	rule
0	Twitter	3000	2584	416	ed_about must equal 1
1	EDTT	85	85	0	retain all public rows

Analysis table dimensions: (2584, 18) and (85, 26)

3. Data description

Observations, purpose, and composition

Twitter: One row represents one sampled post. The analysis table has 2,584 rows and retains the 17 source columns plus a logical-inconsistency flag. id is a synthetic row identifier; the remaining fields are categorical annotations, not symptom severity scores. The release was created to characterize ED/body-image expression. Keyword selection and conditional coding determine which posts and features are represented. Multiple posts may come from one person, but this release cannot identify clusters.

EDTT: One row represents one curated TikTok post (video or photo slideshow). The release was created to study multimodal pro-ED content. The processed table contains 85 rows; its 13 source fields are retained alongside a row ID, 11 header-based

indicators, and a count of recorded EDE categories. Text fields contain summaries and annotations, not the original private transcript. Audio valence and arousal are heuristic musical features, not the emotions of a creator or viewer. The original five numeric fields are tempo in beats per minute, valence, arousal, harmonic ratio, and major/minor mode.

No common clinical scale: Twitter's restraint field is assessed only inside eating-concern posts, whereas EDTT lists restraint as a separate category. Even shared labels do not establish identical measurement. We retain source-specific definitions.

People, consent, and intended use

The Twitter paper reports public posts and an institutional determination that its study did not involve human subjects. That does not show that every poster knew about this specific secondary analysis or consented to it. The EDTT release describes privacy protections and withheld private fields; the inspected documentation does not establish individual notification or consent. We use aggregate summaries, avoid quoting sensitive post summaries, and make no attempts at re-identification. Post annotations cannot diagnose the people who posted them.

Intended use and boundaries

These tables support descriptive secondary analysis and prespecified associations between annotations within each sample. They are unsuitable for diagnosing creators, evaluating moderation decisions about particular accounts, estimating engagement, or ranking platforms by harm. There is no predefined train/test split; no prediction benchmark or model training is performed in Phase II. The analysis-ready files preserve source annotations so a reviewer can trace every derived flag back to its input. Questions about the original annotation or collection should be directed to the source authors rather than inferred from the summaries.

Attribute dictionary

The next tables describe every retained source field and derived variable family. Blank values follow the eligibility or unknown-status rules in Section 2.

Code cell [5]

```
# Present the source-specific dictionary rather than equating similarly named fields.
tw_dictionary = [
 ('id', 'Synthetic unique row ID', 'All 3,000'),
 ('ed_about', '0 unrelated; 1 related; 9 unknown', 'All 3,000'),
 ('shape / eating / weight', 'Concern recorded: 0 no, 1 yes', 'Related posts'),
 ('noconcern', 'None of shape/eating/weight: 1 yes', 'Related posts'),
 ('noimage', '0 image; 1 no image', 'Shape concern'),
 ('body / thighs / stomach', 'Depicted region: 0 no, 1 yes', 'Shape + image'),
 ('hipbones / ribs / collarbones', 'Depicted region: 0 no, 1 yes', 'Shape + image'),
 ('extreme', 'Extreme thinness annotation: 0/1', 'Shape + image'),
 ('restraint / purge', 'Restraint or binge/purge annotation: 0/1', 'Eating concern'),
 ('aversion', 'Aversion to pro-ED content: 0/1', 'Source noconcern=1'),
 ('noconcern_inconsistent', 'Added flag for disagreement with three concern codes', 'Related posts')]
display(pd.DataFrame(tw_dictionary, columns=['Twitter field', 'meaning', 'universe']))
tt_dictionary = [
 ('type', 'video or photos'), ('summary', 'Public descriptive summary'),
 ('ede_subscale_items', 'Recorded EDE category/item text'),
 ('ede_diagnostic_items', 'Recorded diagnostic-feature text; not diagnoses'),
 ('visual_elements / audio_elements', 'Visual annotations / audio description'),
 ('audio_tempo_bpm', 'Audio tempo, beats per minute'),
 ('audio_valence / audio_arousal', 'Heuristic musical positivity / energy, 0–1'),
 ('audio_valence_label / audio_arousal_label', 'low <.33; medium .33–.66; high >.66'),
 ('harmonic_ratio', 'Harmonic sound proportion, 0–1'),
 ('mode_major', '1 major; 0 minor'),
 ('record_id', 'Added sequential row number; not a platform ID'),
 ('ede_shape / ede_eating / ede_weight / ede_restraint', 'Header present=1; absent=0; blank source=NA'),
 ('visual_* (seven indicators)', 'Header present=1; absent=0'),
 ('ede_category_count', 'Number of four headers, 0–4; blank source=NA')]
display(pd.DataFrame(tt_dictionary, columns=['EDTT field', 'meaning']))
```

	Twitter field	meaning	universe
0	id	Synthetic unique row ID	All 3,000
1	ed_about	0 unrelated; 1 related; 9 unknown	All 3,000
2	shape / eating / weight	Concern recorded: 0 no, 1 yes	Related posts
3	noconcern	None of shape/eating/weight: 1 yes	Related posts
4	noimage	0 image; 1 no image	Shape concern
5	body / thighs / stomach	Depicted region: 0 no, 1 yes	Shape + image
6	hipbones / ribs / collarbones	Depicted region: 0 no, 1 yes	Shape + image
7	extreme	Extreme thinness annotation: 0/1	Shape + image
8	restraint / purge	Restraint or binge/purge annotation: 0/1	Eating concern
9	aversion	Aversion to pro-ED content: 0/1	Source noconcern=1
10	noconcern_inconsistent	Added flag for disagreement with three concern...	Related posts

	EDTT field	meaning
0	type	video or photos
1	summary	Public descriptive summary
2	ede_subscale_items	Recorded EDE category/item text
3	ede_diagnostic_items	Recorded diagnostic-feature text; not diagnoses
4	visual_elements / audio_elements	Visual annotations / audio description
5	audio_tempo_bpm	Audio tempo, beats per minute
6	audio_valence / audio_arousal	Heuristic musical positivity / energy, 0–1
7	audio_valence_label / audio_arousal_label	low <.33; medium .33–.66; high >.66
8	harmonic_ratio	Harmonic sound proportion, 0–1
9	mode_major	1 major; 0 minor
10	record_id	Added sequential row number; not a platform ID
11	ede_shape / ede_eating / ede_weight / ede_rest...	Header present=1; absent=0; blank source=NA
12	visual_* (seven indicators)	Header present=1; absent=0
13	ede_category_count	Number of four headers, 0–4; blank source=NA

4. Data limitations and consequences

• **Selection limits the population.** Both collections focus on ED-related material. Twitter keywords and EDTT's curation can overrepresent particular expressions. Their observed percentages are not prevalence estimates for all social media, all eating disorders, or all users. Recommendation surfacing during collection is not a measured user exposure or a randomized algorithm intervention.

• **Platform and period are confounded with collection.** Source size, annotation rules, formats, selection, and time differ. A difference between datasets cannot isolate a platform effect or a historical trend. EDTT post dates are absent from the public file.

• **Unavailable outcomes limit the original question.** Neither supplied CSV has likes, view counts, viewer exposure, longitudinal food attitudes, or clinical outcomes. Thus engagement associations and changes in people's food relationships cannot be estimated here.

• **Annotation is a measurement process.** A category is an annotator's reading of content. The 11 Twitter inconsistencies may affect co-occurrence estimates; without text, we cannot resolve them. Codebook skip rules create structural blanks in Twitter; EDTT's blank EDE fields have an unresolved interpretation. Treating either kind of blank as zero would distort denominators. The EDTT parser extracts categories but cannot repair all ambiguity in free-text descriptions.

• **Dependence and sample size affect inference.** Repeated creators, retweets, or reused audio can make rows dependent. Missing creator identifiers prevent reliable cluster adjustment. With only 85 EDTT rows (79 with EDE text), estimates can be unstable. Statistical intervals in future work will rely on assumptions that the release cannot fully verify.

• **Content is not a diagnosis or a person's state of mind.** Irony, criticism, context, and annotation uncertainty complicate interpretation. A musical valence score is not emotional valence of the message. Equating labels with illness could stigmatize creators or lead to inappropriate moderation.

• **Privacy and amplification matter.** We report aggregates and do not reproduce summaries, usernames, images, or harmful instructions. We cannot verify the expectations of each original poster. Public availability does not erase ethical responsibilities.

These limitations motivate within-source descriptions and cautious association models. A future causal question would require an appropriate exposure/outcome study and its own ethical review and consent process.

5. Exploratory data analysis

5.1 Twitter: theme frequency and conditional denominators

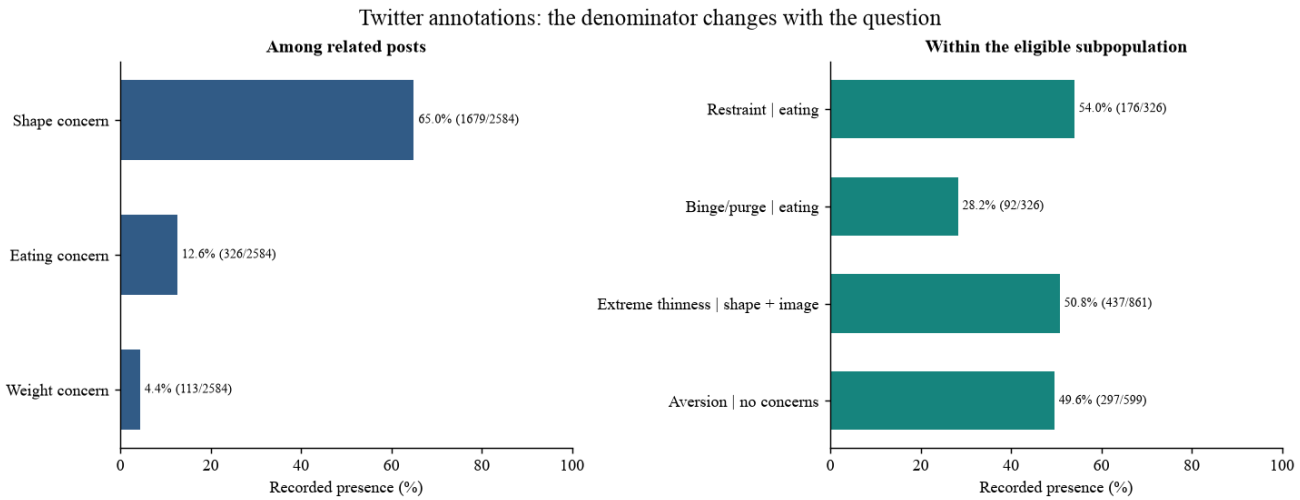
This figure summarizes recorded content, keeping each code's denominator explicit. Shape, eating, and weight can overlap, so their percentages need not sum to 100. Aversion has a different eligible population and is not used as a synonym for recovery. The second panel shows why restraint and visual codes must not be divided by all posts indiscriminately.

These are marginal and nested descriptive summaries. The shape-by-eating association reserved for Statement 1 is not tabulated or modeled.

Code cell [6]

```
# Summarize individual Twitter codes using their codebook-defined denominators.
def prevalence(frame, fields):
    rows = []
    for col, label in fields:
        n = int(frame[col].notna().sum())
        count = int(frame[col].eq(1).sum())
        rows.append([label, count, n, 100 * count / n])
    return pd.DataFrame(rows, columns=['annotation', 'yes n', 'eligible n', 'percent'])
tw_main = prevalence(twitter, [('shape', 'Shape concern'),
                               ('eating', 'Eating concern'), ('weight', 'Weight concern')])
tw_nested = prevalence(twitter, [('restraint', 'Restraint | eating'),
 ('purge', 'Binge/purge | eating'), ('extreme', 'Extreme thinness | shape + image'),
 ('aversion', 'Aversion | no concerns')])
fig, axes = plt.subplots(1, 2, figsize=(11, 4.2), layout='constrained')
for ax, table, color, title in zip(axes, [tw_main, tw_nested], [BLUE, TEAL],
 ['Among related posts', 'Within the eligible subpopulation']):
    ax.barh(table.annotation, table.percent, color=color, height=.6)
    ax.invert_yaxis(); ax.set_xlim(0, 100); ax.set_xlabel('Recorded presence (%)')
    ax.set_title(title, fontsize=11)
    for i, row in table.iterrows():
        ax.text(row.percent + 1, i, f'{row.percent:.1f}% ({row["yes n"]}/{row["eligible n"]})',
                va='center', fontsize=8)
```

```
fig.suptitle('Twitter annotations: the denominator changes with the question', fontsize=14)
finish('01_twitter_themes')
display(pd.concat([tw_main, tw_nested], ignore_index=True).round(1))
display(Markdown('**Interpretation.** Shape concern is the most common of the three main '
 'concerns: 1,679/2,584 (65.0%), versus 326/2,584 (12.6%) for eating and '
 '113/2,584 (4.4%) for weight. Among eating-concern posts, 176/326 (54.0%) '
 'record restraint. This supports retaining body image as a central topic, while '
 'treating food-related concerns as a smaller, explicitly defined subset. These '
 'are annotation frequencies in the selected corpus, not population illness rates.'))
```



	annotation	yes n	eligible n	percent
0	Shape concern	1679	2584	65.0
1	Eating concern	326	2584	12.6
2	Weight concern	113	2584	4.4
3	Restraint | eating	176	326	54.0
4	Binge/purge | eating	92	326	28.2
5	Extreme thinness | shape + image	437	861	50.8
6	Aversion | no concerns	297	599	49.6

Interpretation. Shape concern is the most common of the three main concerns: 1,679/2,584 (65.0%), versus 326/2,584 (12.6%) for eating and 113/2,584 (4.4%) for weight. Among eating-concern posts, 176/326 (54.0%) record restraint. This supports retaining body image as a central topic, while treating food-related concerns as a smaller, explicitly defined subset. These are annotation frequencies in the selected corpus, not population illness rates.

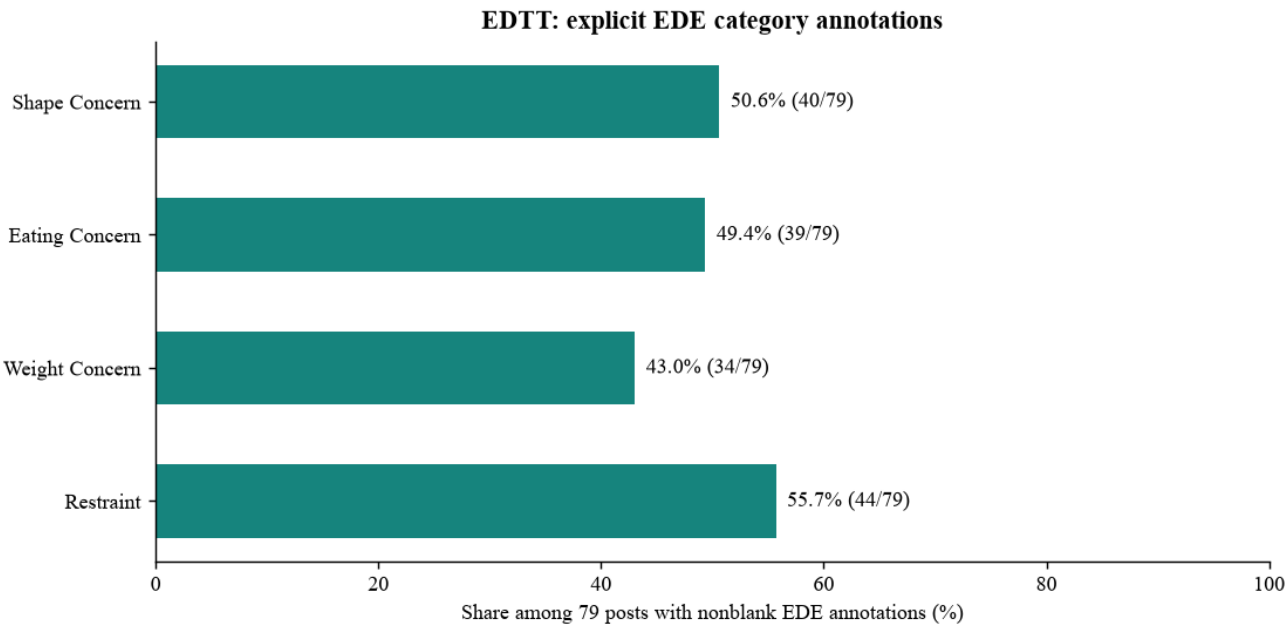
5.2 TikTok: recorded EDE categories and uncertainty from blank annotations

We count explicit category headers. The main percentages use 79 nonblank EDE annotations. For transparency, the table also gives the fraction of all 85 posts with a recorded category and an upper bound if all six blank annotations had that category. This is a missingness sensitivity description, not imputation or a confidence interval. No EDE category is compared to audio valence here.

Code cell [7]

```
# Describe EDE categories and expose the effect of the six blank source fields.
ede_table = prevalence(tiktok, [('ede_' + k, h) for k, h in ede_headers.items()])
```

```
ede_table['all-row lower %'] = 100 * ede_table['yes n'] / len(tiktok)
ede_table['all-row upper %'] = 100 * (ede_table['yes n'] + 6) / len(tiktok)
fig, ax = plt.subplots(figsize=(8.5, 4.1), layout='constrained')
ax.barh(ede_table.annotation, ede_table.percent, color=TEAL, height=.55)
ax.invert_yaxis(); ax.set_xlim(0, 100)
ax.set_xlabel('Share among 79 posts with nonblank EDE annotations (%)')
ax.set_title('EDTT: explicit EDE category annotations')
for i, row in ede_table.iterrows():
    ax.text(row.percent + 1, i, f'{row.percent:.1f}% ({row["yes n"]}/79)', va='center')
finish('02_edtt_ede')
display(ede_table.round(1))
lead = ede_table.sort_values('percent', ascending=False).iloc[0]
display(Markdown(f'**Interpretation.** {lead.annotation} is the most frequently recorded '
 f'category ({int(lead["yes n"])}/79, {lead.percent:.1f}%). Categories can co-occur. '
 'Six blank source fields leave category status unresolved; the all-row bounds span '
 f'{100*6/85:.1f} percentage points. We therefore keep missing flags and avoid '
 'interpreting a blank as evidence that a concern is absent.'))
```



	annotation	yes n	eligible n	percent	all-row lower %	all-row upper %
0	Shape Concern	40	79	50.6	47.1	54.1
1	Eating Concern	39	79	49.4	45.9	52.9
2	Weight Concern	34	79	43.0	40.0	47.1
3	Restraint	44	79	55.7	51.8	58.8

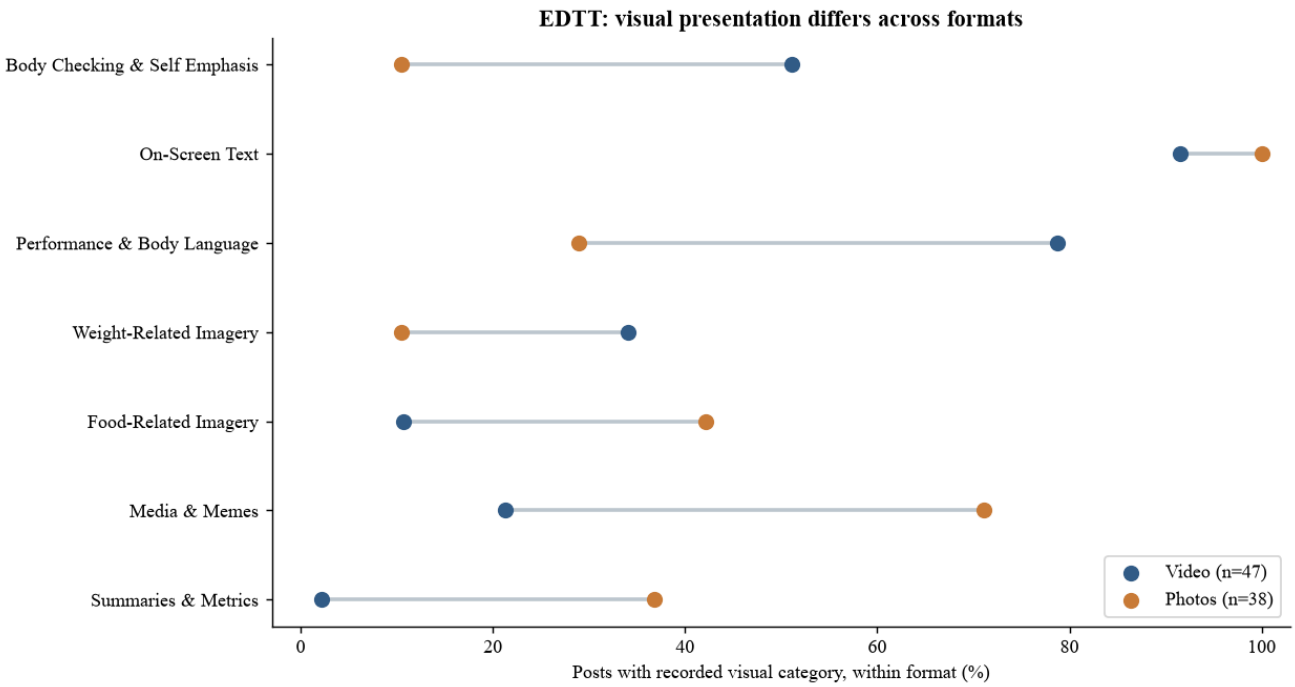
Interpretation. Restraint is the most frequently recorded category (44/79, 55.7%). Categories can co-occur. Six blank source fields leave category status unresolved; the all-row bounds span 7.1 percentage points. We therefore keep missing flags and avoid interpreting a blank as evidence that a concern is absent.

5.3 TikTok: visual annotations by post format

This exploratory pairwise comparison asks whether visual features are recorded differently across videos and photo slideshows. It is separate from both planned analyses. Percentages are calculated within format, so unequal group sizes do not drive the chart. The connected points show descriptive differences only; their lines do not represent time or causation.

Code cell [8]

```
# Explore visual category presence by format without testing a hypothesis.
format_n = tiktok['type'].value_counts()
visual_rates = tiktok.groupby('type')[visual_cols].mean().T.astype(float) * 100
visual_rates.index = list(visual_headers.values())
fig, ax = plt.subplots(figsize=(9.5, 5), layout='constrained')
y = np.arange(len(visual_rates))
for i, row in enumerate(visual_rates.itertuples()):
    ax.plot([visual_rates.iloc[i]['video'], visual_rates.iloc[i]['photos']], [i, i],
            color='#BCC6CE', lw=2, zorder=1)
ax.scatter(visual_rates['video'], y, color=BLUE, s=55,
           label=f'Video (n={format_n["video"]})', zorder=2)
ax.scatter(visual_rates['photos'], y, color=ORANGE, s=55,
           label=f'Photos (n={format_n["photos"]})', zorder=2)
ax.set_yticks(y, visual_rates.index); ax.invert_yaxis(); ax.set_xlim(-3, 103)
ax.set_xlabel('Posts with recorded visual category, within format (%)')
ax.set_title('EDTT: visual presentation differs across formats')
ax.legend(loc='lower right'); finish('03_visual_format')
display(visual_rates.round(1).rename_axis('visual category'))
gap = (visual_rates['photos'] - visual_rates['video'])
label = gap.abs().idxmax()
display(Markdown(f'**Interpretation.** The sample contains {format_n["video"]} videos '
 f'and {format_n["photos"]} photo posts. The largest absolute format difference is '
 f'for {label}: {gap[label]:+.1f} percentage points (photos minus videos). '
 'This suggests format is worth accounting for in the future audio model. '
 'It does not establish that changing format would change content.'))
```



type	photos	video
visual category		
Body Checking & Self Emphasis	10.5	51.1
On-Screen Text	100.0	91.5
Performance & Body Language	28.9	78.7
Weight-Related Imagery	10.5	34.0
Food-Related Imagery	42.1	10.6

type	photos	video
Media & Memes	71.1	21.3
Summaries & Metrics	36.8	2.1

Interpretation. The sample contains 47 videos and 38 photo posts. The largest absolute format difference is for Performance & Body Language: -49.8 percentage points (photos minus videos). This suggests format is worth accounting for in the future audio model. It does not establish that changing format would change content.

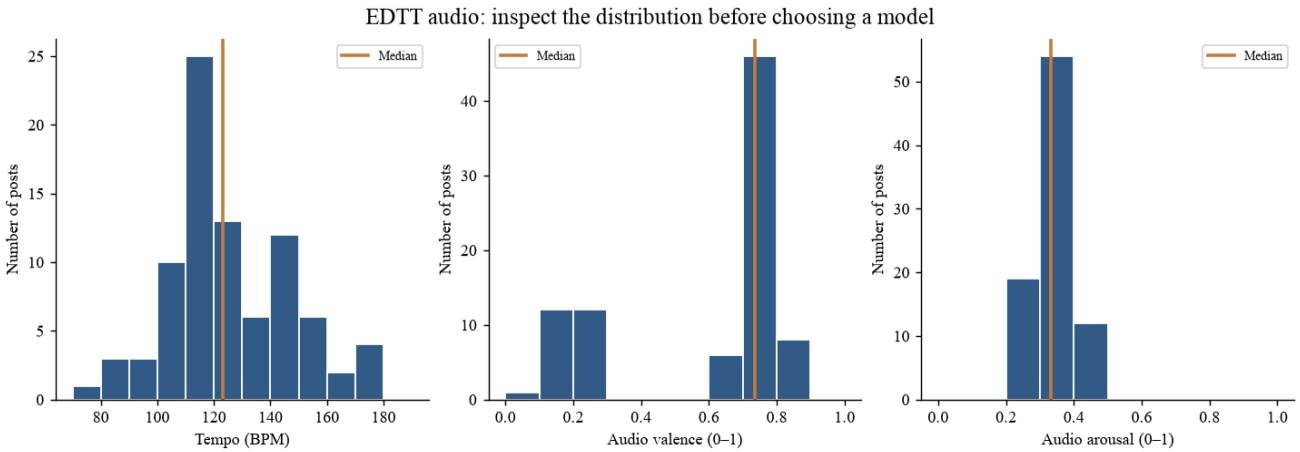
5.4 TikTok: distributions of audio measurements

Histograms describe numeric scale, spread, and shape before modeling. Every released row has these audio fields. The table includes medians and quartiles because averages alone can obscure clusters. The audio descriptors are not measures of audience response. We do not stratify valence by restraint, fit a regression, or compute an association test.

Code cell [9]

```
# Describe audio variables marginally; leave the proposed restraint model untouched.
audio_fields = ['audio_tempo_bpm', 'audio_valence', 'audio_arousal', 'harmonic_ratio']
audio_stats = tiktok[audio_fields].describe().T[['count', 'mean', 'std', 'min', '25%', '50%', '75%',
'max']]
# Display a compact table for the PDF; the complete version is saved as a CSV.
audio_stats.to_csv(CLEAN / 'audio_summary.csv')
display(audio_stats[['count', 'mean', '25%', '50%', '75%']].round(3))
fig, axes = plt.subplots(1, 3, figsize=(11, 3.8), layout='constrained')
for ax, field, label, bins in zip(axes,
 ['audio_tempo_bpm', 'audio_valence', 'audio_arousal'],
 ['Tempo (BPM)', 'Audio valence (0–1)', 'Audio arousal (0–1)'],
 [np.arange(70, 191, 10), np.linspace(0, 1, 11), np.linspace(0, 1, 11)]):
    ax.hist(tiktok[field], bins=bins, color=BLUE, edgecolor='white')
    ax.axvline(tiktok[field].median(), color=ORANGE, lw=2, label='Median')
    ax.set_xlabel(label); ax.set_ylabel('Number of posts'); ax.legend(fontsize=8)
fig.suptitle('EDTT audio: inspect the distribution before choosing a model', fontsize=14)
finish('04_audio_distributions')
display(Markdown(f'**Interpretation.** Median tempo is {tiktok.audio_tempo_bpm.median():.2f} BPM '
 f'(IQR {tiktok.audio_tempo_bpm.quantile(.25):.2f}–{tiktok.audio_tempo_bpm.quantile(.75):.2f}). '
 f'Median valence is {tiktok.audio_valence.median():.3f}, while mean valence is '
 f'{tiktok.audio_valence.mean():.3f}; the histogram shows why one average is an incomplete '
 'summary. Arousal is concentrated in a narrow part of its 0–1 scale. '
 'These distributions motivate robust standard errors and cautious interpretation '
 'of the future mean-difference model, rather than treating audio scores as human emotions.'))
```

	count	mean	25%	50%	75%
audio_tempo_bpm	85.0	126.039	112.350	123.050	143.550
audio_valence	85.0	0.590	0.232	0.736	0.774
audio_arousal	85.0	0.342	0.303	0.332	0.385
harmonic_ratio	85.0	0.742	0.670	0.789	0.865



Interpretation. Median tempo is 123.05 BPM (IQR 112.35–143.55). Median valence is 0.736, while mean valence is 0.590; the histogram shows why one average is an incomplete summary. Arousal is concentrated in a narrow part of its 0–1 scale. These distributions motivate robust standard errors and cautious interpretation of the future mean-difference model, rather than treating audio scores as human emotions.

5.5 What the EDA changes

We retain food and body-image content as the shared motivation but replace the original algorithm/engagement question with answerable within-source questions. The Twitter sample has a large shape-concern group and a smaller eating-concern group. EDTT contains both post formats and multimodal features, but only 79 records have interpretable EDE category status. This motivates one Twitter association model and one EDTT audio model, with prespecified missing-data rules.

The four figures were selected to show category frequencies, the impact of conditional denominators, one distinct exploratory pairwise relationship, and numeric distributions. No inferential test, fitted model, p-value, or preregistered predictor–outcome cross-tabulation has been run.

6. Questions for reviewers: Part 1

The main issues we want feedback on are whether our annotations measure the ideas in our question, how missing and inconsistent codes affect interpretation, and whether the two sources form a coherent project.

1. **Unresolved Twitter annotations:** Eleven posts have all three concern codes set to zero but noconcern=0. Is preserving them with a flag and checking the future model both with and without them preferable to excluding them now?

2. **Missing TikTok annotations:** Six EDTT rows have blank EDE text. Is treating their category status as unknown reasonable, and should we seek author clarification before deciding on any alternative coding?

3. **Different sampling and definitions:** Does presenting Twitter and TikTok as complementary case studies sufficiently connect the project, given that their percentages cannot identify platform differences and restraint has different coding rules?

4. **Construct validity:** Our original interest included food attitudes and recovery, but the files measure content annotations. Does the revised question stay close enough to our topic while avoiding unsupported claims about people or algorithm effects?

5. **EDA priorities:** Is the visual-feature comparison by format useful preparation for the planned audio model, or would you prioritize a different descriptive view without testing our preregistered associations?

Part 2. Preregistration of future analyses

Status and disclosure. These statements specify future work after the Phase II data audit and EDA. They are not a claim that the data have never been viewed or that an external registry entry already exists. Source documentation, published aggregate results, row examples, a three-concern consistency check (including the 11 inconsistent rows), marginal distributions, and the format–visual comparison were inspected. The two proposed predictor–outcome associations below were not tabulated, tested, or modeled. Save a dated copy of these plans before performing them; document any later amendments and reasons.

Reporting commitment. The final report will include both analyses regardless of direction or statistical significance, along with sample sizes, exclusions, effect estimates, uncertainty, diagnostics, and limitations. The two primary tests form one family: use two-sided p-values and Holm correction at family-wise alpha 0.05 (smaller p compared with .025, then the other with .05 if the first passes). Report unadjusted and adjusted p-values. Statistical significance will not substitute for substantive interpretation.

7. Preregistration Statement 1: Co-occurring concerns on Twitter

Hypothesis. Among posts coded as ED/body-image related, shape-concern annotations are positively associated with eating-concern annotations. The rationale is that body-related and food-related themes can be expressed together; this is a content-level conjecture, not a claim about illness or causation. The design permits a negative or null result.

Population and variables. Use twitter_related.csv with ed_about=1 (currently 2,584 posts). Outcome eating is 1 if eating concern was coded and 0 otherwise. Predictor shape is 1 if shape concern was coded and 0 otherwise. Use these original source codes without recoding restraint into the outcome. Exclude any row with a missing predictor/outcome, report the count, and do not impute. The current validation finds none. Keep all valid IDs, including the 11 flagged rows, because there is no evidence identifying an error in shape or eating specifically; do not deduplicate code patterns. Report a prespecified sensitivity refit excluding the 11 flagged rows, clearly secondary to the primary all-row model. The logical audit exposed their zero codes before this plan, which is disclosed rather than treated as unseen data.

Model and estimand. Fit a logistic regression:

$$\text{logit}[P(\text{eating} = 1)] = \beta_0 + \beta_1 * \text{shape}$$

The target is the within-sample odds ratio $\exp(\beta_1)$ comparing shape-annotated with non-shape-annotated posts. Report both group probabilities as well, so an odds ratio is not mistaken for a risk ratio. The directional expectation is $\beta_1 > 0$, but the primary test is two-sided $H0: \beta_1 = 0$.

Procedure, uncertainty, and diagnostics. In the final phase only, report the 2-by-2 table, fit the model with statsmodels, and use the two-sided coefficient test with its 95% confidence interval. Report raw cell counts and flag any count below five. If any cell is zero or maximum likelihood fails due to separation/non-convergence, use a two-sided Fisher exact test as the prespecified fallback, report the empirical odds ratio (including zero/infinity if applicable), and obtain an exact conditional 95% interval with scipy.stats.contingency.odds_ratio. Do not add arbitrary pseudo-counts or search for a better-significance specification. If a variable has no variation, report that the association is not estimable.

Interpretation and limits. The model describes annotation co-occurrence in this selected corpus. It does not imply that body-image concerns cause eating concerns. Shared annotation practices, repeated creators, and retweets could affect the association and precision. Without creator IDs, independent-row standard errors may be too small; acknowledge this rather than claiming population generalizability. We do not add noconcern, which is determined by the concerns, as a control.

What will be shown later: counts, group probabilities, odds ratio and interval, test result, any fallback, and a short explanation even if the estimate contradicts the hypothesis.

8. Preregistration Statement 2: Restraint annotations and audio valence in EDTT

Hypothesis. Posts with a recorded restraint category have lower mean audio valence than posts without that category, conditional on format. This is a tentative hypothesis about the musical framing of food restriction. The link between semantic content and musical tone is uncertain; ironic or upbeat accompaniment could produce the opposite pattern. No association result is being assumed.

Population and variables. Use EDTT rows with a nonblank ede_subscale_items field and valid valence and format (currently 79 eligible rows). Outcome is audio_valence on its released 0–1 scale. Predictor ede_restraint is 1 when the explicit Restraint header appears and 0 when the nonblank field lacks it. photos is an indicator for type='photos'; video is the reference. Exclude blank EDE fields from the primary analysis and report six currently excluded rows. Do not treat the diagnostic field as a clinical outcome, remove outliers, or transform valence after seeing the coefficient.

Model and estimand. Fit ordinary least squares:

audio_valence = beta_0 + beta_1 * ede_restraint + beta_2 * photos + error

The target is the adjusted mean difference in valence units, beta_1. The expectation is beta_1 < 0, with a two-sided primary test H0: beta_1 = 0. Format is specified in advance because audiovisual construction may differ across formats. This model meets the requirement for a model relating variables to the research question; it does not make a causal adjustment claim.

Procedure and uncertainty. Fit with statsmodels and HC3 heteroskedasticity-robust standard errors using its t-based inference option (use_t=True). Report coefficient, 95% interval, primary p-value, eligible group sizes, adjusted predictions by restraint and format, and R-squared. Inspect residual-versus-fitted values, leverage, and Cook's distance. Flag Cook's distance above 4/n but retain those rows in the primary model. Report predictions outside 0–1 if they occur. If the design matrix is rank deficient or restraint has no variation, report non-estimability rather than choosing a different predictor.

Prespecified sensitivity, not a replacement primary test. Refit the same model twice including the six blank-EDE rows: once coding their restraint status as 0 and once as 1. Report the resulting coefficients and intervals as extreme-assignment scenarios, not as sharp bounds or as evidence that missingness is random. Keep the original complete-annotation model as primary regardless of significance. These sensitivities are not additional confirmatory claims.

Interpretation and limits. A difference in heuristic musical valence is not evidence of a difference in viewers' emotions, symptoms, or engagement. HC3 addresses unequal residual variance, not dependence from reused audio or creators. The purposive sample and modest size restrict generalization and power. Diagnostics are reported transparently, not used to select a more favorable hypothesis.

What will be shown later: sample flow, the regression table, an adjusted-prediction plot, diagnostics, sensitivity estimates, and interpretation even when the interval includes zero.

Appendix A. Data cleaning appendix and reproducibility

Files. data/raw/ contains unchanged source downloads and the checksum manifest. data/intermediate/ contains validated/normalized snapshots. data/processed/ contains analysis tables and audits. figures/ contains the four charts. The download bundle includes these files, the executed notebook, its HTML view, and the combined PDF. In GitHub, the notebook is in notebooks/ and the report is in reports/phase2/.

Re-run. Extract the full ZIP, install the packages in requirements.txt in a Python environment, open this notebook from the extracted directory, select that environment's kernel, and choose Restart Kernel and Run All. No network access or collection code is needed during execution. Processed files and figures are overwritten reproducibly. All code cells should finish without errors. No proposed inferential analyses are implemented here.

Hosted raw inputs. The verified GitHub raw-data folder contains both original CSVs, the Twitter codebook and filtering appendix, and manifest.json. These are the exact bytes used in this execution. The publisher links in Section 2 provide independent

access to the original releases.

Cleaning trace. Section 2.2 verifies the four original files; Section 2.3 validates Twitter codes and their conditional universes, saves all 3,000 rows, retains 2,584 related posts, and flags 11 logical inconsistencies; Section 2.4 normalizes all 85 EDTT rows and derives 11 category indicators plus a count while preserving six unknown EDE annotations; Section 2.5 saves row-flow and eligibility audits. Sections 5.1–5.4 generate the displayed summaries and figures from those analysis-ready files. Every code cell has an executed output or visible saved-file result, and the proposed inference remains unperformed.

Appendix B. References

1. Cavazos-Rehg, P. A., et al. (2019). “I just want to be skinny.”: A content analysis of tweets expressing eating disorder symptoms. *PLOS ONE*, 14(1), e0207506. https://doi.org/10.1371/journal.pone.0207506. S1 Dataset, S1 Codebook, and S1 Appendix are bundled unchanged.

2. Shaveet, E., Viranda, T., & Choudhury, T. (2026). *EDTT Dataset*, version 2. Zenodo. https://doi.org/10.5281/zenodo.19749827. Public dictionary: https://edenshaveet.github.io/EDTT/.

3. Project repository: https://github.com/AnyaChoudhary/info2950-fall2026-project. Inspection snapshot recorded in Section 2.

4. INFO 2950, Fall 2026. *Phase II: Exploratory analysis and hypotheses*. User-supplied course rubric, read in full. Its datasheet-style prompts guide Sections 3 and 4.

9. Questions for reviewers: Part 2

For the final phase, we want feedback on whether our proposed models answer meaningful content questions and whether their uncertainty and sensitivity plans are adequate for these samples.

1. **Model choice:** Is logistic regression for Twitter concern co-occurrence appropriate at this course level, with Fisher's exact test as the prespecified fallback for zero cells or separation?

2. **Audio interpretation:** Is musical valence a meaningful way to study content framing, provided we do not interpret it as a creator's or viewer's emotional state? Should the hypothesis be reframed before any model is fit?

3. **Bounded outcome and small sample:** Is a mean-difference model with HC3 uncertainty appropriate for valence among 79 EDTT posts? If a different method is preferable, what estimand should a dated amendment preserve?

4. **Missingness sensitivity:** Are assigning all six unknown restraint labels to zero and then to one useful scenario checks, given that these are not sharp bounds? Should source-author clarification take priority?

5. **Dependence and uncertainty:** Without reliable creator or repeated-audio identifiers, how prominently should we qualify the independence assumption, and should the report emphasize descriptive effect estimates over significance?

6. **Scope and coherence:** Do the two within-source analyses adequately address one project about food and body-image content, or should we narrow the final report while still presenting both preregistered analyses regardless of results?